

# Colby Frison

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## EDUCATION

### University of Oklahoma, Gallogly College of Engineering

Bachelor of Science in Computer Science. President's Honor Roll (Spring 2025). GPA: 3.76/4.00.

Norman, OK

Expected December 2026

Relevant Coursework: Compiler Construction, Data Structures & Algorithms, Software Engineering, Computer Organization, Programming Languages

## SKILLS

**Languages:** Python, TypeScript, JavaScript, C++, C, Java, HTML/CSS, PowerShell

**Technologies:** React, Next.js, Node.js, Express, Firebase, MongoDB, Tailwind CSS, Git, GitHub, CI/CD, Power Automate Desktop, UiPath

**Concepts:** Full-Stack Development, Compiler Design, AST Transformations, SSA, RPA, Process Automation, Machine Learning

## EXPERIENCE

### IT Intern, Automation & AI Development

May 2026 – July 2026

ConocoPhillips

Bartlesville, OK

- Evaluated platform gaps between Power Automate and existing RPA tools to support enterprise adoption and integration
- Served as sole developer on migration of 2 UiPath workflows into Power Automate Desktop pilot projects, documenting findings and delivering reports to senior stakeholders to guide platform decisions
- Documented capability differences across RPA platforms to inform evaluation and adoption planning for the Automation & AI group

### Engineering Ambassador

March 2025 – August 2025

University of Oklahoma, Gallogly College of Engineering

Norman, OK

- Served as sole Computer Science ambassador, communicating complex technical concepts to prospective students and families
- Led campus tours of engineering facilities and provided detailed information about majors and departmental opportunities
- Supported Engineering Days and high school camps through workshop guidance, event operations, and student engagement

### Teacher / Teacher's Assistant

March 2020 – May 2024

Houston Museum of Natural Science

Sugar Land, TX

- Led and managed summer camp classes independently in the instructor's absence, demonstrating leadership and adaptability
- Developed lesson plans and educational materials for science-focused programs
- Mentored students and junior staff to foster an engaging learning environment

## PROJECTS

### Tensor Compiler Optimization Pipeline

Fall 2025

Compiler Construction Project, University of Oklahoma

- Built a schedule-driven pipeline transforming YAML-defined tensor kernels through AST lowering passes into executable C
- Implemented hoisting, modulo simplification, pointer-dereference lowering, and kernel-to-SSA passes to improve code quality
- Created CLI verification and benchmarking workflows to validate correctness and compare optimization variants reproducibly

### Classroom Q&A

March 2025 – May 2025

Personal/Academic Project

- Developed and deployed a full-stack anonymous Q&A app adopted by roughly 150 students in a live university course
- Implemented real-time sync with Firebase/Firestore, TypeScript, React, Next.js, and Tailwind CSS; increased engagement prompted other students to build competing systems

### FreshStart (2nd Place, Hacklahoma 2024)

February 2024

Hacklahoma

- Achieved 2nd place out of 200+ participants while leading front-end development for an award-winning farm planning web app
- Built interactive farm planning UI with MongoDB, Express, React, and Node.js (MERN stack)
- Integrated machine learning for crop yield prediction and an LLM for personalized farm task generation

## ACTIVITIES & LEADERSHIP

### Pride of Oklahoma Marching Band

August 2023 – December 2024

Member

- Balanced a rigorous rehearsal and performance schedule with full-time coursework under high-pressure game-day conditions